

DRT and The Future of Mobility

A Model for Equitable Transport



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About Me

- From Nebraska
- Learned to Drive in Montana
- Studied in Germany
- Settled in Spain
- Learned about Mobility as an Adult







Representatives from Jeju Island and [@embcoreaespain](#) visited the CENIT offices today to discuss smart transport projects in Korea and Europe.



What is DRT?

DRT = Demand Responsive Transport

- Shared ride system
- Dynamic routing
- Paratransit/ Special needs
- Operates in a zone
- Dispatch can be real-time or scheduled ahead
- Between fixed-route transit and taxi/ride-hail
- Operational efficiency



Positioning

Where DRT fits in the mobility spectrum



- Best for medium-to-low demand where fixed routes are inefficient
- Works as a feeder to high capacity trunk lines
- Service quality must be maintained to avoid “slow shared taxi” perception



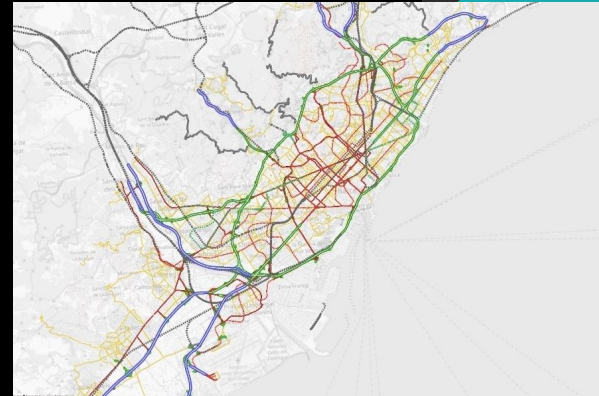
Government considerations for DRT

Infrastructure

- No extra infrastructure needed
- Needs vehicle stock
- Route development can be AI optimized
- Payment system

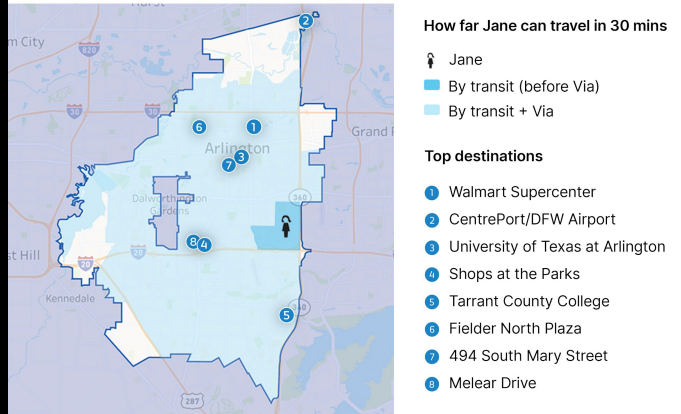
Smart City Integration

- Tool for equity building
- Mobility plan completion
- Adaptable solution
- Great solution for groups with reduced mobility



Government considerations for DRT

Overview of Arlington service zone.

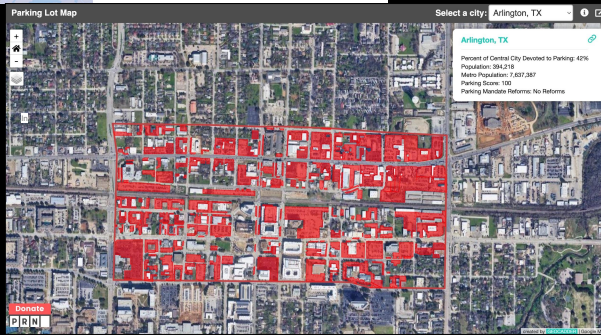


Route Development

- Can augment existing public transportation network
- Can be implemented in lower density areas
- Servicing based on demand

Environmental

- Advantages of zero carbon fuels
- AI-optimization
- Enables communal transport
- Reduces need for parking spaces



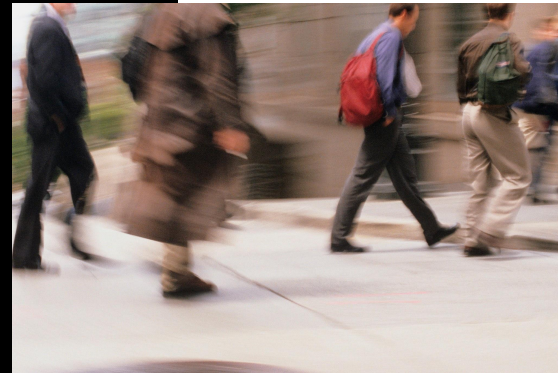
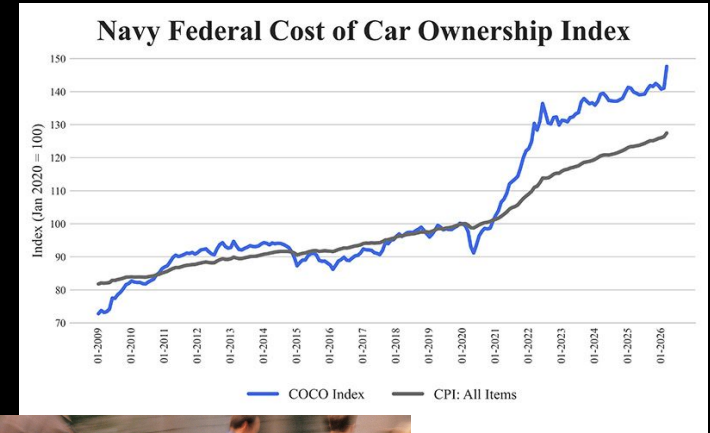
Transport is the way people access opportunities

Financial

- Life without private vehicle
- No Insurance
- No parking
- Potential subsidies

Health

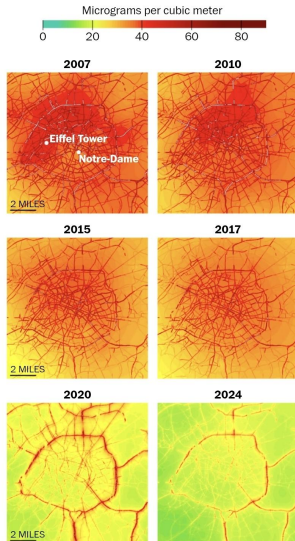
- Enables community interaction
- Enables movement
- Decreases stress of driving and responsibility of car ownership
- Mental Health



Transport is the way people access opportunities



Average nitrogen dioxide concentration in Paris



Source: Airparif

Community

- Enables transport across population segments
- Transport 4 All
- Passive sharing of space
- Enables connection

Environmental

- Reduces pollution
- Potential increase in green space
- Supports increased sustainable mobility planning

Case Study VARO DRT- Transport for Aging Well

★ Cheongju, South Korea

A solution for the aging population
(Problem)

- Inefficiencies and rural blind spots
- Long waiting times
- Unsafe driving
- Physical isolation
- Last-Mile disconnect



Case Study VARO DRT- Transport for Aging Well

★ Cheongju, South Korea

A solution for the aging population
(Results)

- Waiting time decreased by 83% (from 110 min to less than 10 min)
- Smaller buses and AI gave the city more operational efficiency
- Multi-channel access
- 50+% increase in monthly ridership





Development of Korea's First Integrated Mobility Platform for Optimized Operation of All Transport Modes

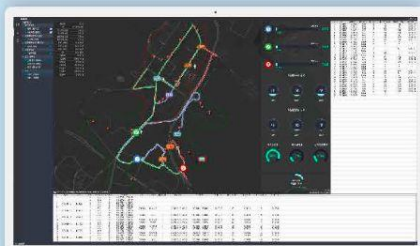
TAMOS

First in Korea

Integrated Mobility Platform based on AI & Big Data: Planning – Operation – Monitoring

Planning

Designing Optimal
Operation Models
for Service Areas



Operation

Driver Terminal



User's
Hailing Methods

App



Call
Center

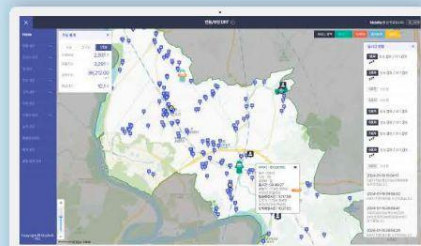


Smart
Bell



Monitoring

Real-time Service Monitoring
/ Data Disclosure

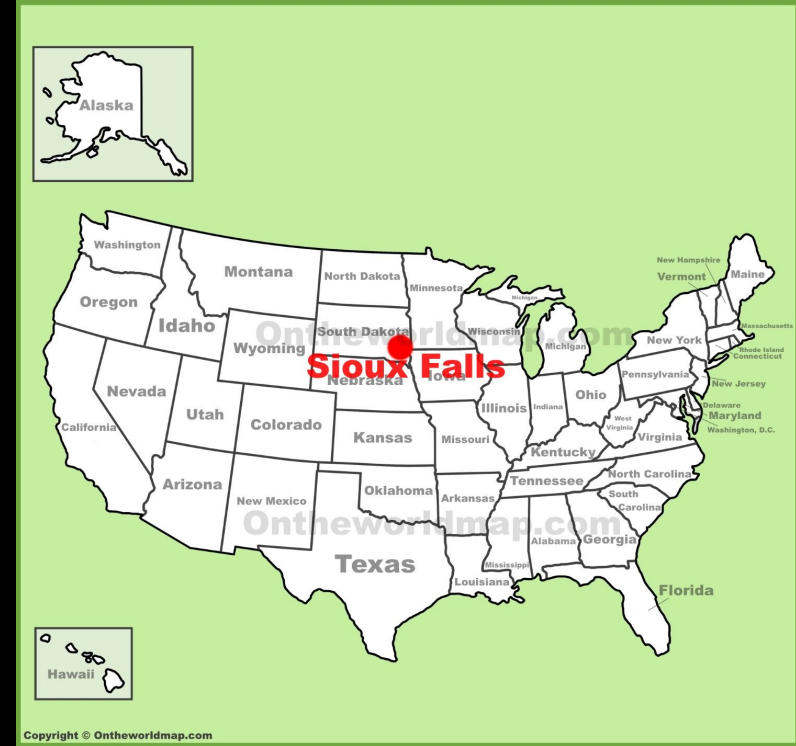


Case Study SAM on Demand-Transport for Access

★ Sioux Falls, South Dakota

Solution for growth (problem)

- Low density city
- >50% growth rate over last 20 years
- Fixed-route, non-optimized transport
- Facing a financial cliff in the city budget



Case Study SAM on Demand-Transport for Access

★ Sioux Falls, South Dakota

Solution for growth (results)

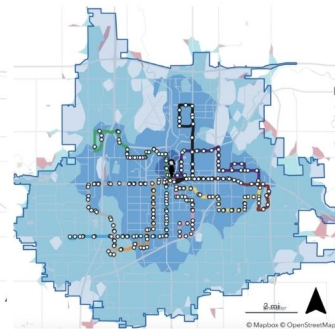
- Partnered with Via for DRT service, quality of Life approach
- 100% unionized operator staff
- 100% coverage throughout city
- 2x job access within 1 hour
- 25% reduction in cost per ride

Before



(Source: Via Remix,
Sioux Falls Map)

After



(Source: Via Remix,
Planning tool)



Thank you!

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